Flowcharts_and_Prompts_Guide

Graph Digital Twin Project Guide

August 4, 2026

Contents

Guide: Project Flowcharts & AI Diagram Prompts

This guide provides both **Mermaid.js code** (for rendering diagrams directly in Markdown/Notion) and **Miro AI Prompts** (to generate interactive, editable diagrams in Miro).

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Flowchart 1: The Graph Digital Twin Security Pipeline

This diagram explains the step-by-step processing of grid telemetry through the Graph Digital Twin, GNN model, and explainability engine.

1. Mermaid.js Code

Copy this block into any Markdown viewer (GitHub, Notion, Obsidian) to render the flowchart:

```
graph TD
    %% Define Styles
    classDef physical fill:#e1f5fe,stroke:#0288d1,stroke-width:2px;
    classDef processing fill:#fff3e0,stroke:#f57c00,stroke-width:2px;
    classDef decision fill:#e8f5e9,stroke:#388e3c,stroke-width:2px;
    classDef output fill:#f3e5f5,stroke:#7b1fa2,stroke-width:2px;

    %% Nodes
    A[SCADA Grid Telemetry: V, Theta, P, Q] --> B[Impedance Matrix: Resistance R, Reactance
    B --> C[Graph Construction: Nodes=Buses, Edges=Lines]:::physical
    C --> D[Multi-Task GraphSAGE Backbone]:::processing
    D --> E{Attack Flagged?}:::decision

    E -- No --> F[Normal Operation: Logging]:::output
    E -- Yes --> G[Trigger Localization Head]:::processing

    G --> H[Flag Compromised Substations]:::output
    H --> I[Integrated Gradients XAI Engine]:::processing
    I --> J[Generate Feature Attribution Heatmaps]:::output

    %% Style Assignment
    class A,B physical;
```

2. Miro AI Prompt

Copy and paste this prompt into Miro AI (Command: "/generate flowchart" or use the Miro AI App) to create the diagram:

¿ "Generate a professional technical flowchart for a Graph Digital Twin smart grid cybersecurity pipeline. Start with SCADA Telemetry (Voltage Magnitude V, Angle Theta, Active Power P, Reactive Power Q) and Grid Impedance parameters (R, X) feeding into 'Graph Construction'. From Graph Construction, point to 'Multi-Task GraphSAGE Backbone'. Then, branch into a decision block named 'Attack Detected?'. If No, point to 'Normal Operation'. If Yes, point to 'Substation Localization Head' which identifies compromised nodes. Finally,

point the Localization Head to 'Integrated Gradients Explainability (XAI) Engine' to generate node-level attribution maps for utility operators."

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Flowchart 2: The Closed-Loop Cyber-Mitigation Loop

This diagram explains the active grid healing loop once the GDT flags a substation as hijacked.

1. Mermaid.js Code

Copy this block into any Markdown viewer to render the flowchart:

```
graph TD
    %% Define Styles
    classDef detection fill:#ffebee,stroke:#c62828,stroke-width:2px;
    classDef action fill:#fff3e0,stroke:#f57c00,stroke-width:2px;
    classDef restoration fill:#e8f5e9,stroke:#388e3c,stroke-width:2px;

    %% Nodes
    A[GNN Localization Head Flags compromised Bus 5]:::detection --> B[Substation Cyber-Isol
    B --> C[Mute SCADA Telemetry at Bus 5]:::action
    C --> D[Load Historical Baseline Profile]:::action
    D --> E[Inject +-2% Forecasting Uncertainty]:::action
    E --> F[Substitute Telemetry with Healthy Baseline]:::restoration
    F --> G[Run State Estimator on Restored Grid Profile]:::restoration
    G --> H[WLS Error drops from 0.007228 to 0.000002 p.u.^2]:::restoration
    H --> I[Grid Tracking Recovered by 99.97%]:::restoration

    %% Style Assignment
```

2. Miro AI Prompt

Copy and paste this prompt into Miro AI to create the diagram:

¿ "Create a workflow diagram representing a Closed-Loop Cyber-Physical Mitigation (Grid Healing) system for electrical substations. The flow should start with 'GNN Flags Compromised Nodes' (e.g. Bus 5). Point to 'Trigger Cyber-Isolation'. From there, split into two actions: 'Mute Corrupted Sensor Telemetry' and 'Fetch Historical Load/Generation Profiles'. Point both actions to 'Inject 2% Forecast Uncertainty'. This leads to 'Substitute Bad Telemetry with Baseline Estimates'. Next, connect to 'Execute AC State Estimator'. End the flowchart with a success node: 'Grid State Tracking Restored (WLS MSE drops by 99.97%)'."